Roll Number: _____

Thapar Institute of Engineering and Technology (TIET), Patiala

Department of Computer Science & Engineering (CSED)

MST EXAMINATION

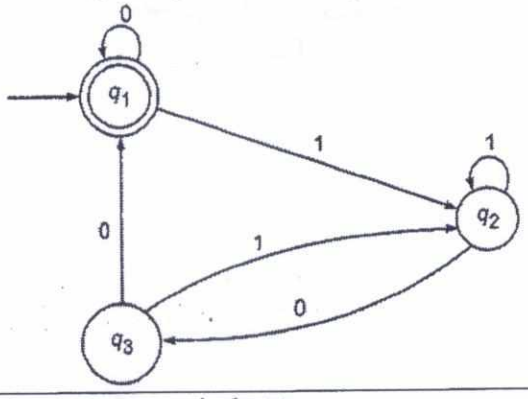
B. E. (3rd Year)	Course Code: UCS701/UCT301 Course Name: Theory of Computation/ Formal Language and Automata Theory
01st July 2025	Tuesday, 03:00 PM - 05:00 PM
Time: 02:00 Hrs., Max. Marks: 30	Name of Faculty: Dr. Shailendra Tiwari

Note: Attempt all questions. Assume missing data, if any suitably.

1. Construct minimal DFA that accepts all string of a's and b's where: [4]

$$n_a(w) \cong 0 \; mod \; 2 \;\; \&\& \;\; n_a(w) \cong 0 \; mod \; 2$$

2. Construct a regular expression corresponding to the state diagram described by Fig. below [4]



3. Construct a deterministic finite automaton equivalent to [4]

$$M = (\{q_0, q_1, q_2, q_3\}, \{0, 1\}, \delta, q_0, \{q_3\})$$

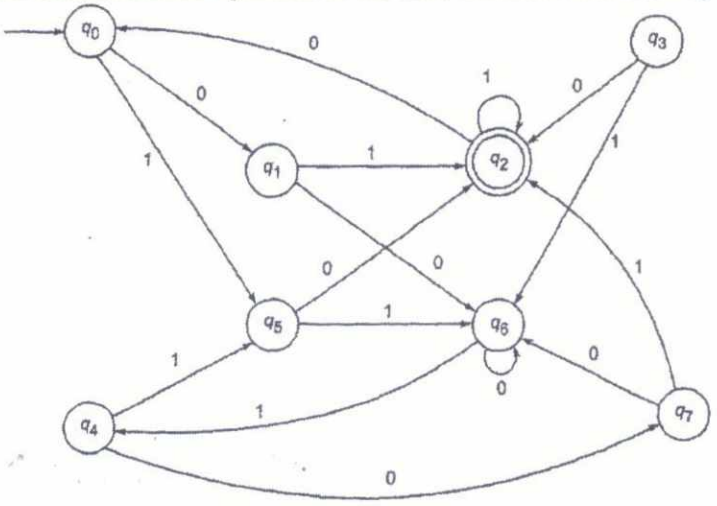
Where δ is given by Table:

State/Σ	a	b
→q_0	q_0, q_1	q_0
q_1	q_2	q_1
q_2	q_3	q_3
(q_3)		q_2

4. Write a left-linear and right-linear regular grammar over $\sum = \{a, b\}$, such that string contain at most three a's. [3]

5. Consider the Mealy machine described by the transition table given by Table below. Construct a Moore machine which is equivalent to the Mealy machine. [5]

Present state	Next state			
	Input a = 0		Input a = 1	
	state	output	state	output
→q_1	q_3	0	q_2	0
q_2	q_1	1	q_4	0
q_3	q_2	1	q_1	1
q_4	q_4	1	q_3	0

6.	Use Thompson's construction to construct a NFA for the following regular expression $(aa \mid b)^*(a \mid cc)^*$ and convert this NDFA into DFA by using subset construction and then minimize it.	[5]
7.	Construct a minimum state automaton equivalent to the finite automaton described by the Fig.1 below  Fig.1	[5]

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